

# Molecular Simulations of Thermosetting Polymers

Cameron F. Abrams

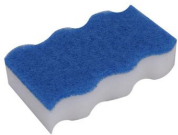
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Virginia Tech – Chemical Engineering  
Blacksburg, Virginia – 12 October 2026



# Thermosetting Polymers: Extraordinary Everyday Materials

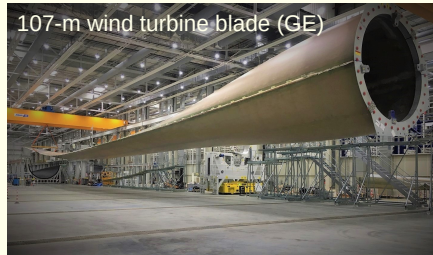
Melamine “sponge”



Solid-fuel ramjet missile  
(Norway)



107-m wind turbine blade (GE)



Adhesives



Vintage bakelite billiard balls



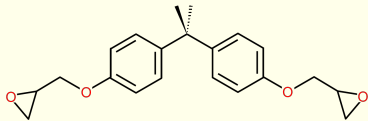
Boeing 787 Dreamliner (Wikipedia)



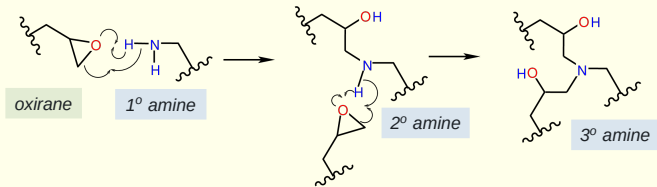
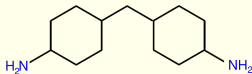
# Common Thermosets: Epoxy Resins and Amine Hardeners

## Examples:

Diglycidyl ether of bisphenol A (DGEBA)



4,4'-diaminodicyclohexylmethane (PACM)



# Thermosetting Polymer Design and Use Considerations

Strong and light

Easily processed  
*Low-viscosity  
prepolymer*

Hard to recycle

strength  
  
toughness

Corrosion/  
chemical  
resistant

Insulating

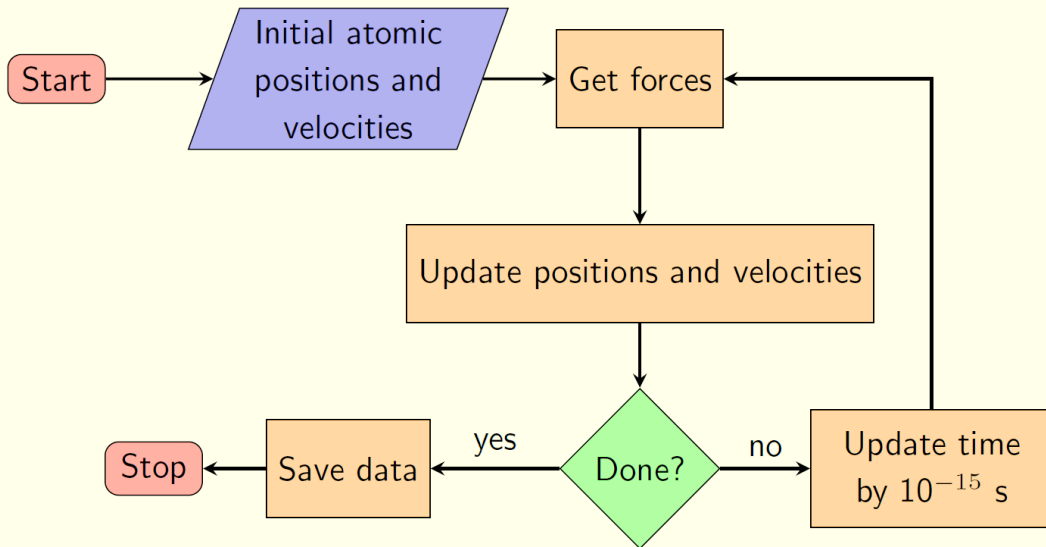
Rely on  
petrochemical  
precursors

thermomechanical  
stability  
  
toughness

A key imperative: Improve *toughness* without sacrificing strength and thermomechanical stability

What generic topological features determine thermoset toughness, and how can we realize them in actual systems?

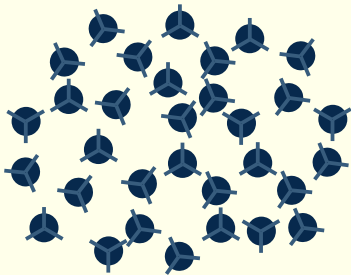
# Molecular Dynamics (MD) Simulations



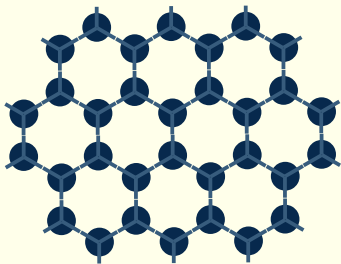
# Early work: Generic Thermosets and 3D Network Regularity

Uniform bond valence angles;  $z = 3$

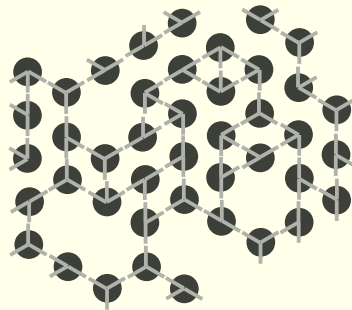
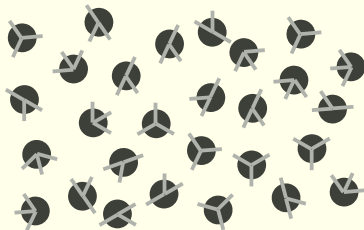
Liquid



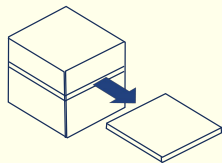
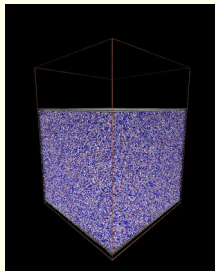
Network polymer



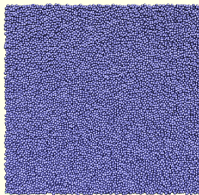
Nonuniform bond valence angles ;  $z = 3$



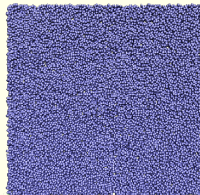
# Protovoid-based strain hardening of generic network polymers



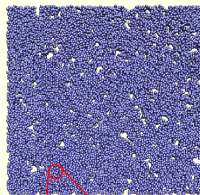
510,000 LJ particles



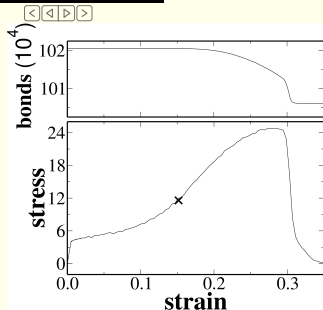
liquid



95% crosslinked  
 $z=4$

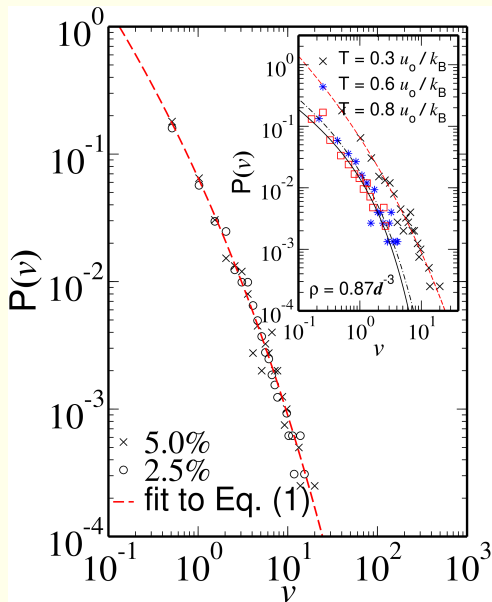


5% tensile strain  
 $\dot{\epsilon} = 2 \times 10^{-4} \tau^{-1}$



Voids open before bonds start breaking

# Void sizes conform to equilibrium free-volume theory



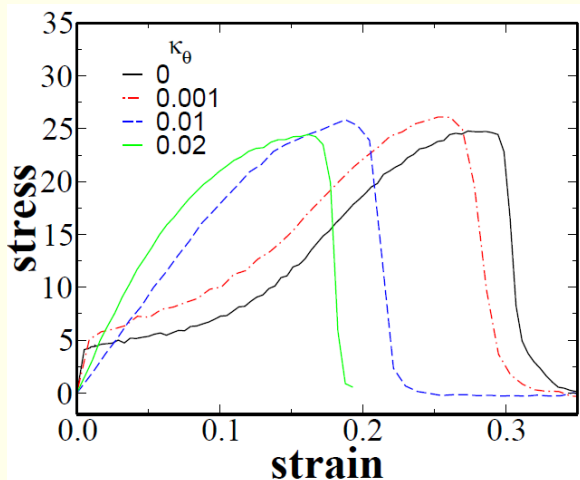
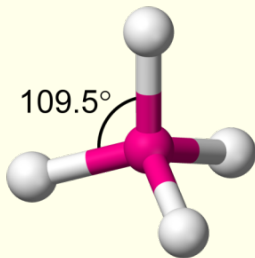
$$P(v) \propto v^{\alpha-1} \exp\left(-\frac{v^\zeta}{k_B T}\right)$$

Hoover, Hoover & Hanson. J Chem Phys 1979;70:1837

- Protovoids are encoded by bonding geometry
- Protovoids are a “natural” part of a system’s void spectrum
- Strain provides energy to actualize protovoids

# Voids can be suppressed using constrained bond angles

$$V_{\text{angle}}(\theta_{ijk}) = \kappa_{\theta} (\theta_{ijk} - \theta_0)^2$$

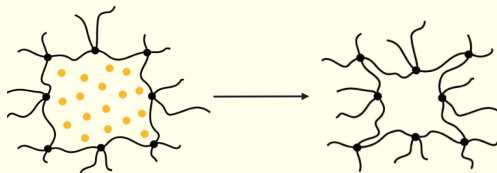


## Protovoid: a definition

- A void encoded by bonding geometry *before* any strain is applied
- Latent free volume that strain converts into real void space
- Suppressed by constraining bond angles — therefore designable

# Engineering protovoid-based toughening into real epoxy thermosets

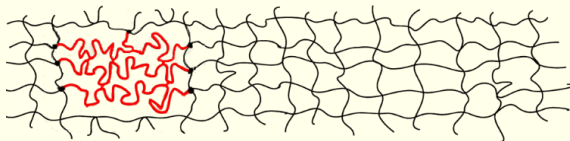
## Reactive encapsulation of solvent and drying (RESD)



Sharifi, Jang, Abrams, Palmese. J Mater Chem A 2014;2:16071

Jang, Sharifi, Palmese, Abrams. Polymer 2014;55:3859

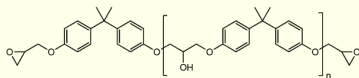
## Partially reacted substructures (PRS)



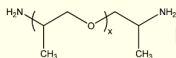
Sharifi, Jang, Abrams, Palmese. Macromolecules 2015;48:7495

Jang, Sharifi, Palmese, Abrams. Polymer 2016;90:249

# Reactive encapsulation of solvent and drying (RESO): Experiments



Diglycidyl Ether of Bisphenol-A (DGEBA), EPON 828,  $n=0.13$



Polyetheramine (Jeffamine D400),  $x=6.1$

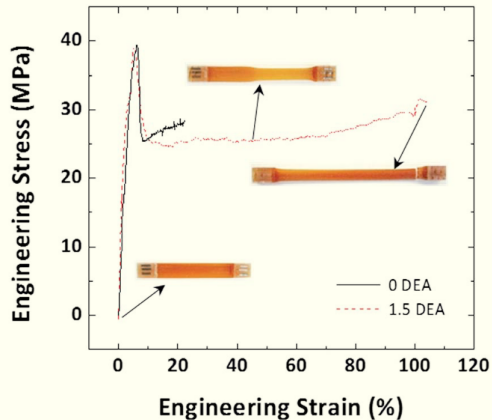
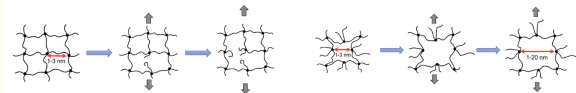
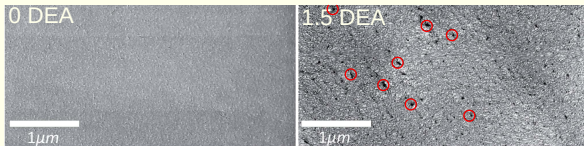
Low  $T_g$



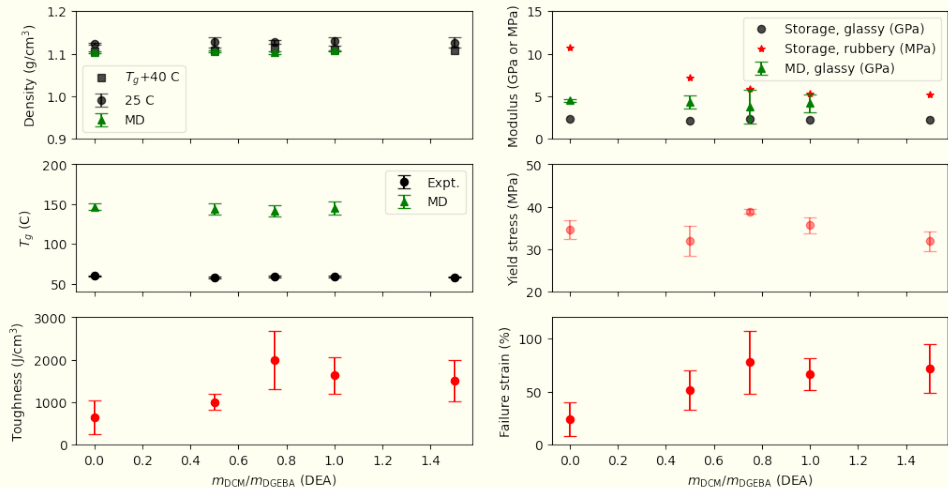
Dichloromethane (DCM)

$$DEA = \frac{m_{DCM}}{m_{DGEBA}}$$

Fracture surfaces after tensile failure



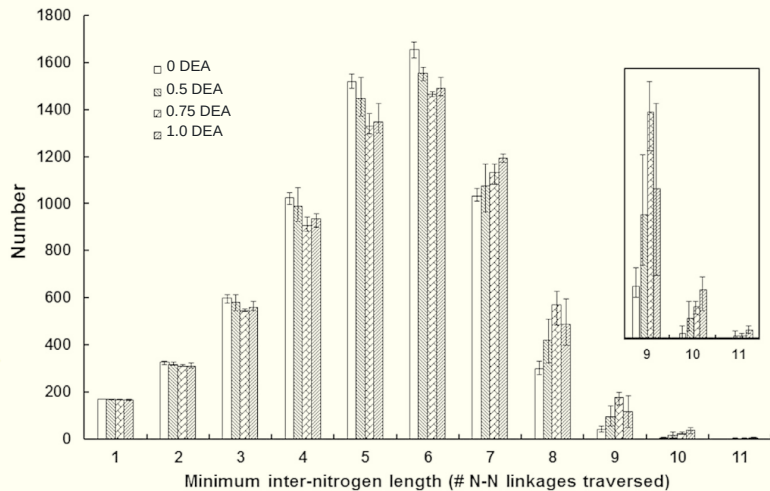
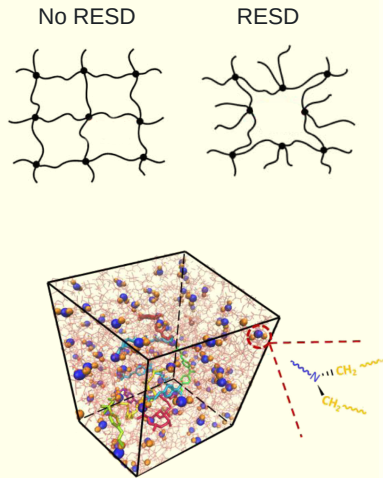
# RESO: Solvent effects partially reproduced by MD



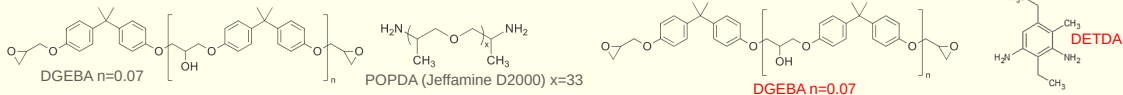
Sharifi, Jang, Abrams, Palmese. J Mater Chem A 2014;2:16071

Jang, Sharifi, Palmese, Abrams. Polymer 2014;55:3859

# MD-simulated RESD network architecture is consistent with voids

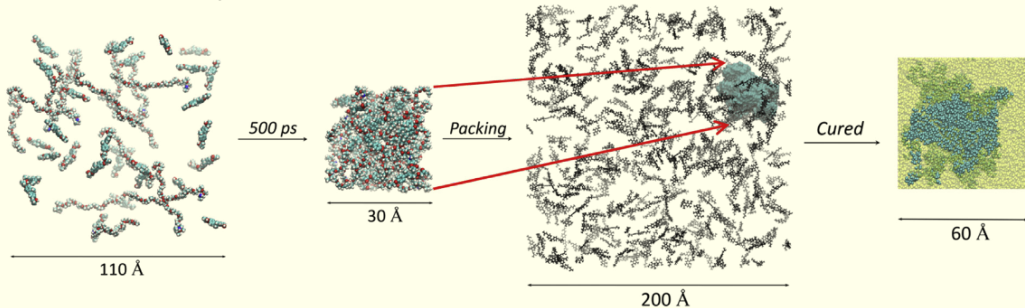


# Partially reacted substructures (PRS)



Partial curing of low- $T_g$  network into a PRS

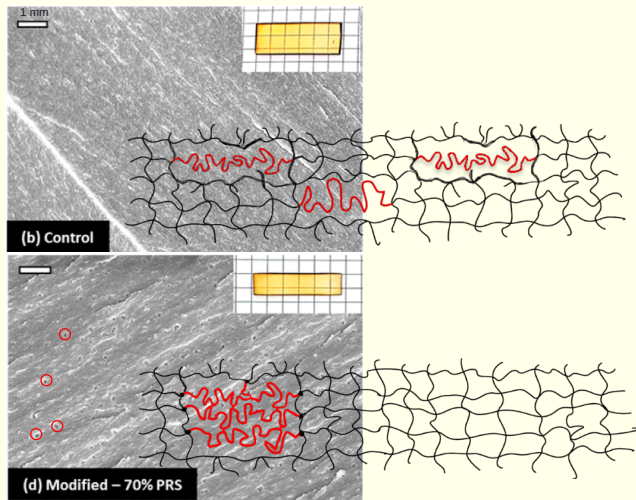
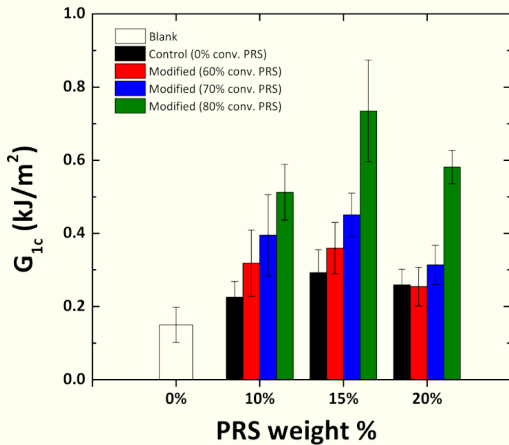
Curing of **high- $T_g$**  network matrix with embedded PRS



Variables: (1) Degree of PRS cure and (2) wt % of PRS in composite system

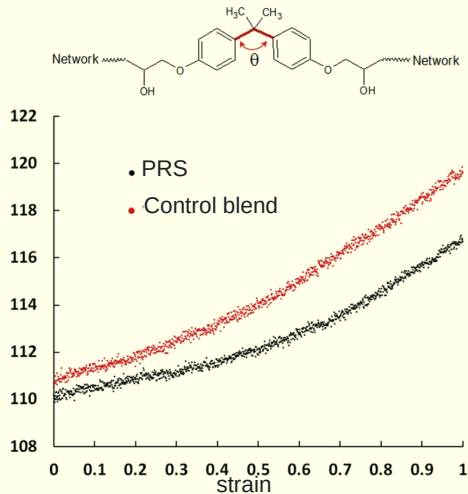
# PRS: Protovoid-induced toughness enhancement

$G_{1c}$ : strain energy release rate

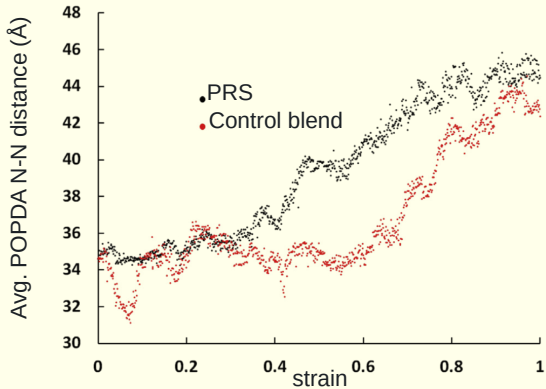


# PRS: MD simulations of 15-wt% of 60% cured PRS vs. control blend

Avg. DGEBA central C-C-C angle (degrees)

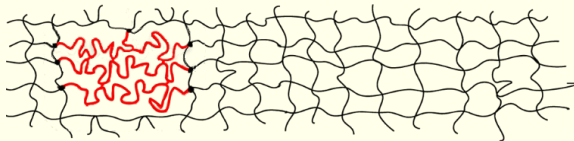
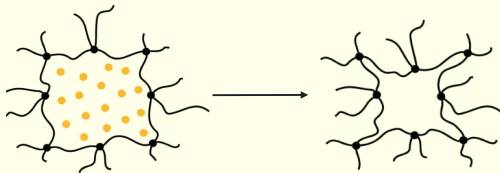


Angle-loading is delayed in PRS



PRS yields as POPDA's stretch cooperatively

## The story so far...



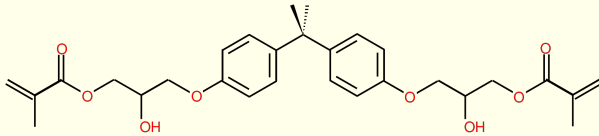
Control of network architecture is key for improving toughness without sacrificing modulus and thermal stability ( $T_g$ )

So far, only specialized blends have been used...

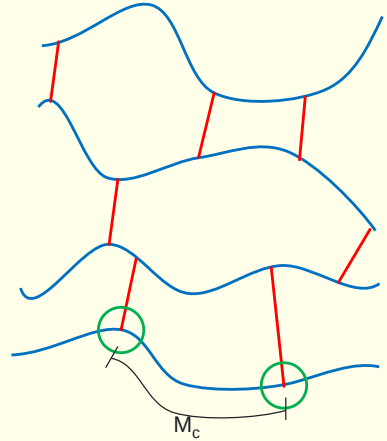
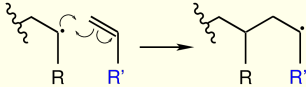
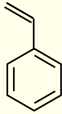
# Common Thermosets: Vinyl Ester Resins and Styrene Diluents

Examples:

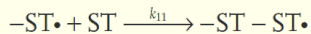
Bisphenol A-glycidyl methacrylate (bis-GMA) "VE"



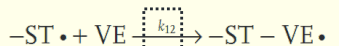
Styrene "ST"



# Reactivity ratios in vinyl-ester/styrene thermosets



$$k \propto p$$



Assumed equal

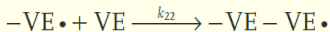


$$r_1 = \frac{k_{11}}{k_{12}} = \frac{p_{11}}{p_{12}}$$

$$p_{11} = r_1 p_{12}$$

$$r_2 = \frac{k_{22}}{k_{21}} = \frac{p_{22}}{p_{21}}$$

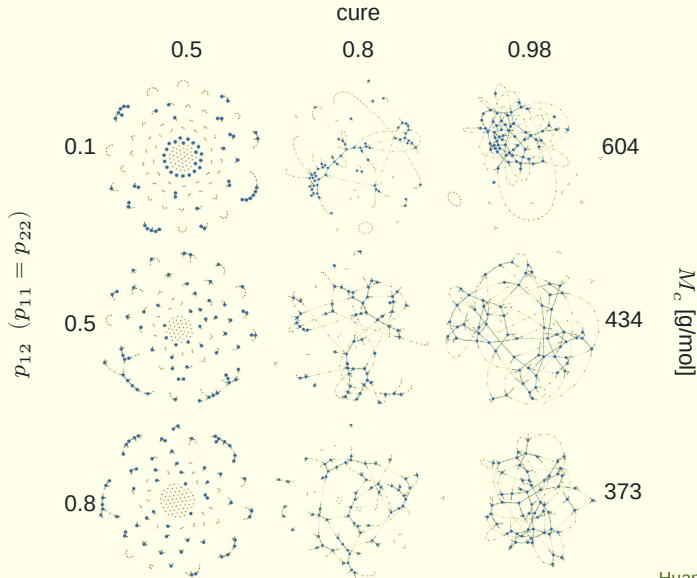
$$p_{22} = r_2 p_{12}$$



$$p_{11} + p_{22} + p_{12} = 1$$

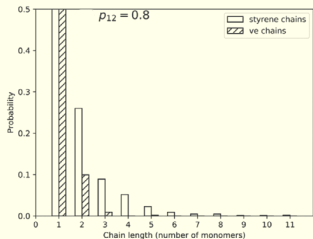
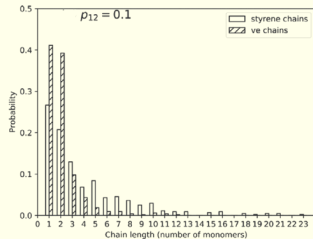
| Temperature [°C]  | $r_1$ | $r_2$ | $p_{11}$ | $p_{22}$ | $p_{12}$ |
|-------------------|-------|-------|----------|----------|----------|
| 30 <sup>a)</sup>  | 0.45  | 1.54  | 0.15     | 0.52     | 0.33     |
| 90 <sup>a)</sup>  | 0.47  | 0.61  | 0.23     | 0.29     | 0.48     |
| 90 <sup>b)</sup>  | 0.49  | 0.5   | 0.25     | 0.25     | 0.50     |
| 140 <sup>c)</sup> | 0.36  | 0.24  | 0.23     | 0.14     | 0.63     |

# Network topology: sensitivity to $p_{12}$ at constant VE:ST



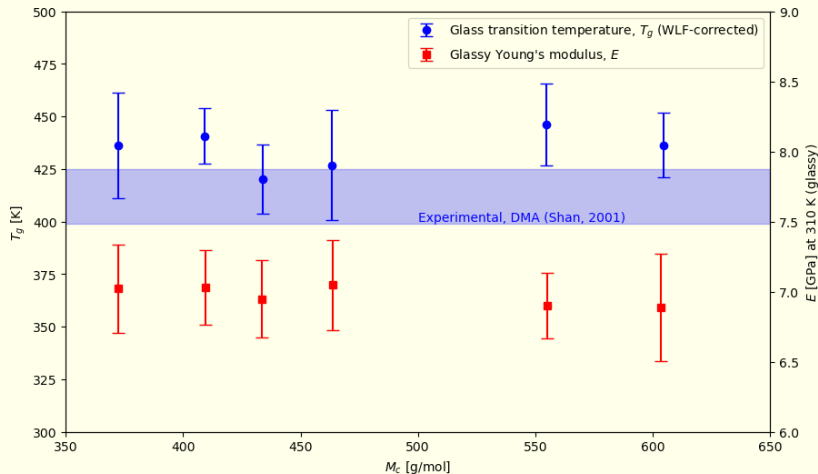
- Higher  $p_{12}$  means more VE-ST cross-propagation.
- $M_c$  falls from 604 to 373 g/mol at fixed VE:ST.
- Composition alone does not set topology.

# Chain-length distributions shift sharply with $p_{12}$



- At  $p_{12} = 0.1$ , styrene homopropagates into long chains — a distribution with a heavy tail out past 20 monomers.
- At  $p_{12} = 0.8$ , cross-propagation truncates everything: almost nothing exceeds 5 monomers.
- Same VE:ST ratio in both panels. Only the reactivity ratio changed.

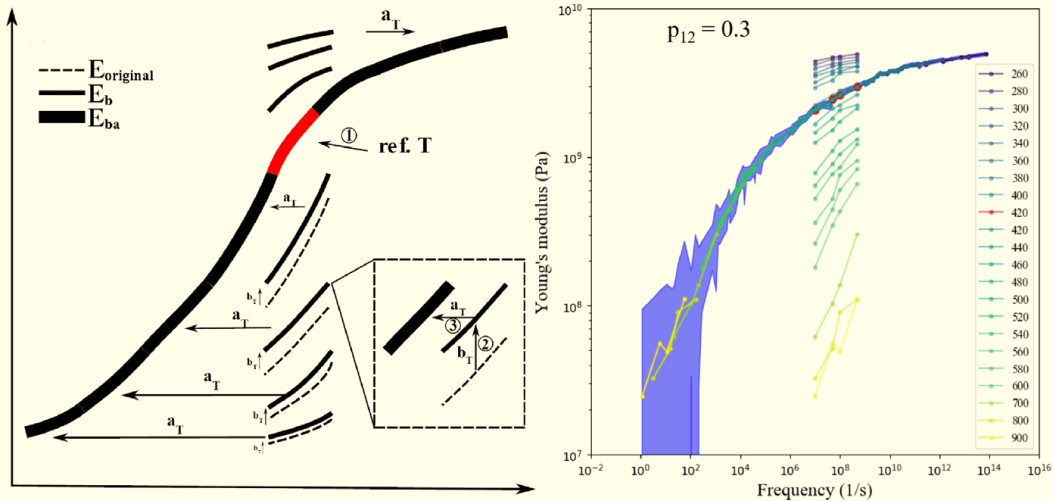
# Network topology: insensitivity of $T_g$ and $E$ at constant VE:ST



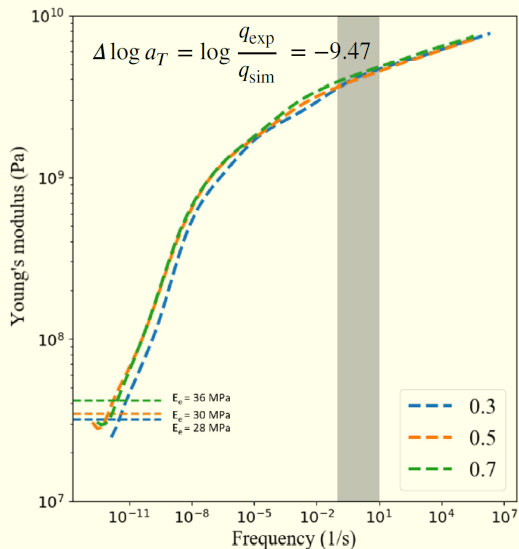
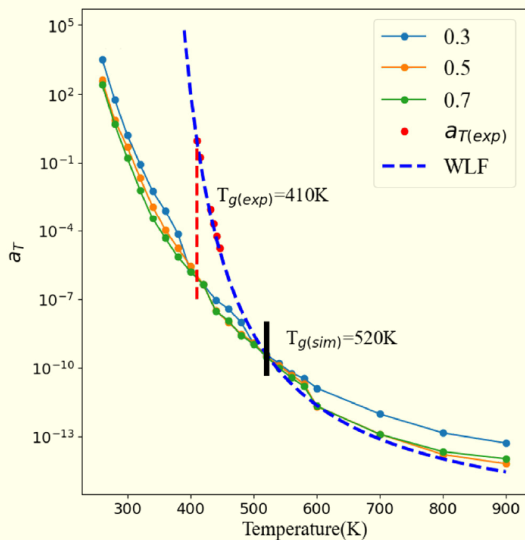
WLF corrects  $T_g$  for heating-rate discrepancy

Young's modulus is a factor of 2 too high!

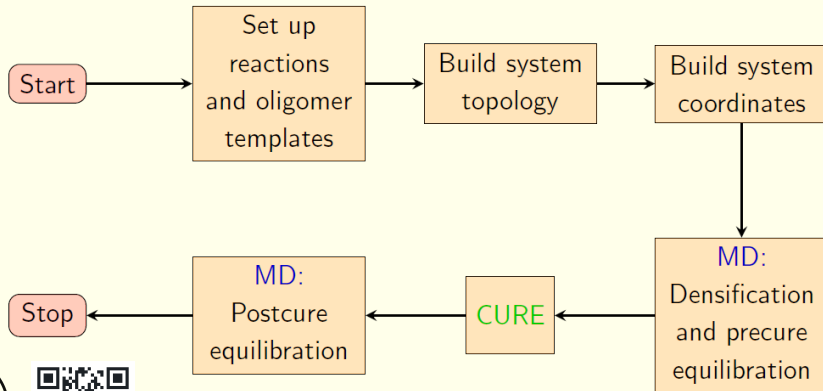
# Estimating quasi-static E using time-temperature superposition



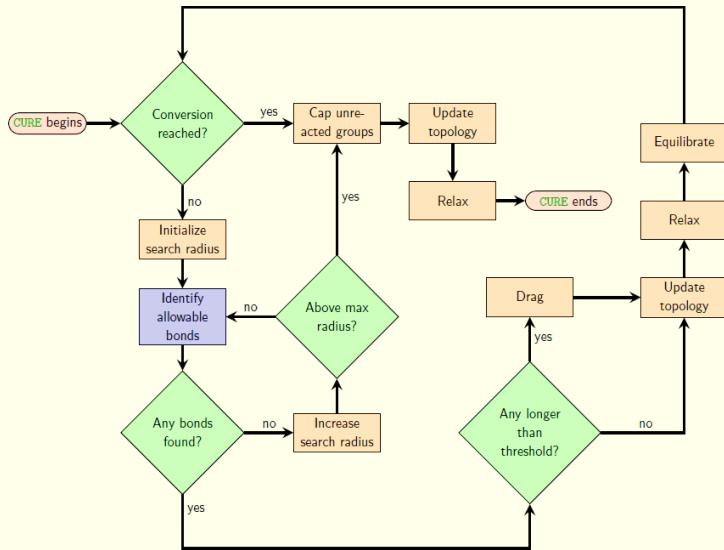
# TTSP with cooling-rate correction gives correct glassy modulus



# htpolynet: Network Polymer MD System Generator



# htpolynet: The CURE Algorithm



# htpolynet example: DGEBA/PACM (inputs)

User input: one YAML file



DGEBA

DGE:

```
smiles: "CC(C)(c1ccc(OC[CH:3]([OH:5])[CH3:1])cc1)c1ccc(OC[CH:4]([OH:6])[CH3:2])cc1"
reactive_atoms: {1: C1, 2: C2, 3: C3, 4: C4, 5: O1, 6: O2}
count: 200
```



PACM

PAC:

```
smiles: "C1C[CH:1](CCC1CC2CC[CH:2](CC2)[NH2:3])[NH2:4]"
reactive_atoms: {1: C1, 2: C1, 3: N1, 4: N1}
count: 100
```



Chemistry  
+ config.

reactions:

- name: Primary-to-secondary-amine  
reactants: {1: PAC, 2: DGE}  
product: PAC~N1-C1~DGE
- name: Secondary-to-tertiary-amine  
reactants: {1: PAC~N1-C1~DGE, 2: DGE}  
product: PAC~N1-C1~DGE-C1~DGE

CURE:

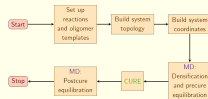
controls:

```
search_radius: 0.5 # nm
desired_conversion: 0.95
```

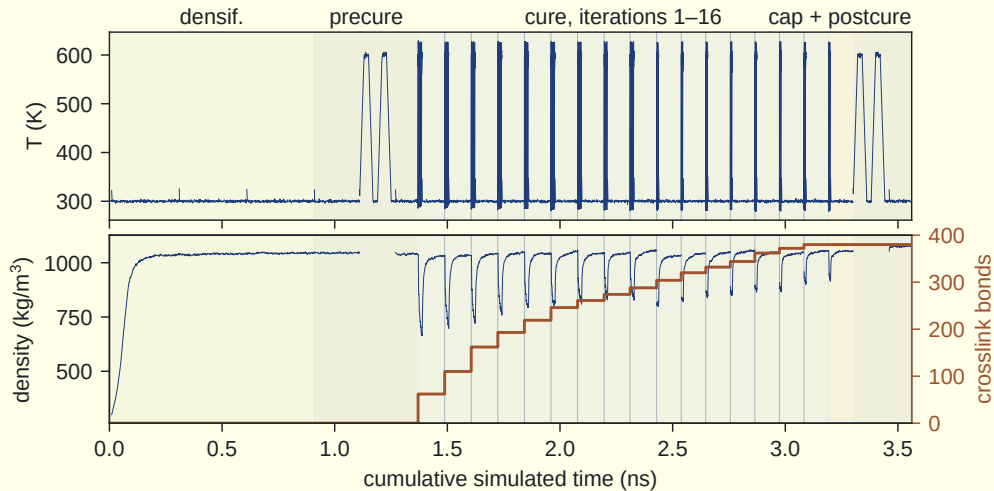
densification: ...

precure: ... postcure: ...

Monomers are built from atom-mapped SMILES (the map numbers mark reactive atoms).  
htpolynet generates every oligomer template it needs from the two reactions and their symmetry equivalents, parameterizes them with GAFF, then densifies, cures and equilibrates.

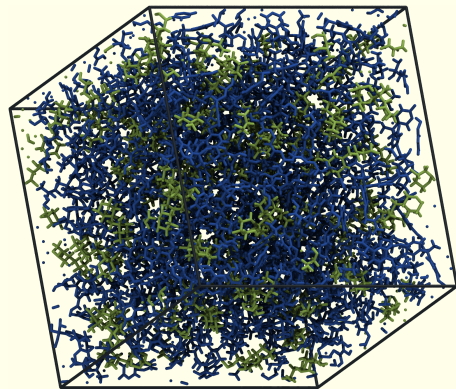
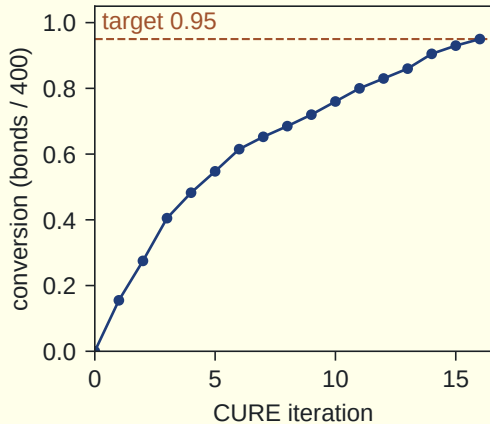


# htpolynet output: the build, stage by stage



Example 3: 200 DGEBA + 100 PACM, htpolynet 2.8. 16 cure iterations reach 380 of 400 possible crosslinks (0.95). The 600 K spikes are the relax stages inside each iteration.

# htpolynet output: cure conversion and the final network



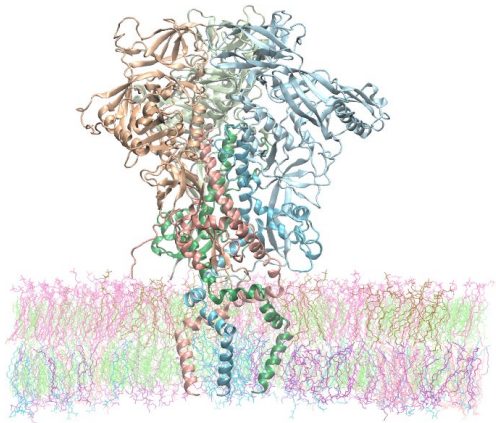
box edge 5.14 nm

■ DGEBA ■ PACM

Final density at 300 K: 1076 kg/m<sup>3</sup> (100 ps NPT average).  
One network of 13,802 atoms, plus 2 unreacted DGEBA.

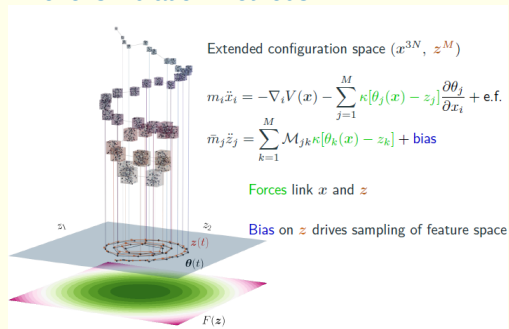
# Other recent projects

## Modeling the pretriggered HIV-1 spike

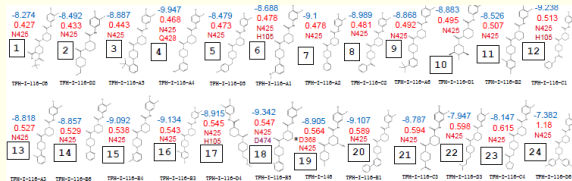


(built using pestifer [<https://pestifer.readthedocs.io/en/latest/>])

## Novel simulation methods: TAMM



## Computational Drug Design



## Key takeaways

- Thermosets can be toughened without sacrificing thermomechanics via *protovoids*
- Protovoid-based toughening demonstrated in generic MD simulations and several experimental systems
- Protovoids are a class of generic topological features that can be leveraged to implement new toughening approaches
- Network topology can be controlled independently of monomer composition via reactivity ratios

# Acknowledgments

Those who did the work discussed in this talk:

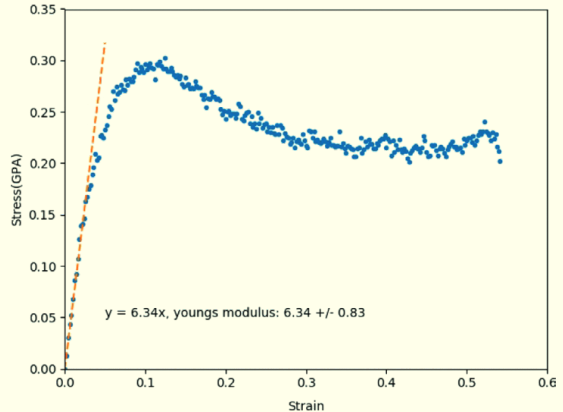
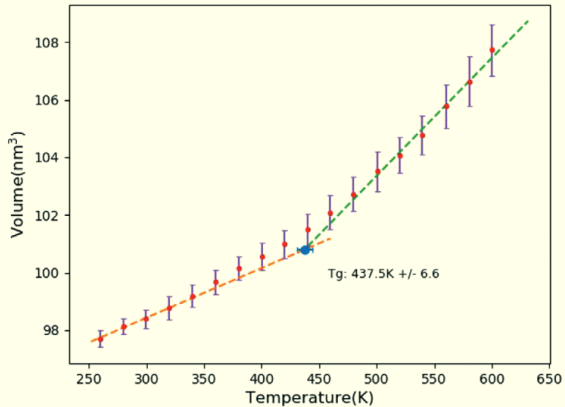
- Debashish Mukherji
- Changwoon Jang
- Majid Sharifi
- Ming Huang
- Arun Srikanth
- Giuseppe Palmese

Funding for the work discussed in this talk:

- US Army W911NF-06-2-0013, W911NF-07-1-0301, W911NF-12-2-0022, W911NF-12-R-0011, W911NF-13-2-0046, W911NF-17-2-0227
- NSF AST-0959884
- ACS PRF 42368-G7

Backup slides

# Examples of measuring Tg and E (glassy)



## WLF-correction to $T_g$

$$\Delta T_g = \frac{-C_2 \log_{10} \frac{q_{\text{exp}}}{q_{\text{sim}}}}{C_1 + \log_{10} \frac{q_{\text{exp}}}{q_{\text{sim}}}} \quad (C_1 = 17.3, C_2 = 92.4) \text{ with } q_{\text{exp}} = 3.33 \text{ K/s and } q_{\text{sim}} = 10^{10} \text{ K/s}$$